Solution to Sequences Puzzle

Liz Burnell

1. 16 (Multiply by 2)
2. 48 (Primes $2, 3, 5, 7, 11, 13$ plus 1)
3. S (Saturday)
4. D (December)
5. 18 (Lucas sequence)
6. .... (Morse code: $M =$ --, $A =$.-, $T =$ -, so $H =$ ....)
7. 8 (Digits of π)
8. `Pro Scientia Atque Sapientia` (Stuyvesant motto)
9. 8 (Digits of Euler's number e)
10. 62 ($x^2 - 2$)
11. 0 (Arithmetic modulo 7)
12. 10 (Pascal's triangle read by rows)
13. $-i$ (Powers of i)
14. 312211 (Look-and-say sequence)
15. 4 (Number of letters in month names)
16. 397 (Centered hexagonal numbers)
17. 235416 (Sum of first n cubes; $(1 + \cdots + n)^2$)
18. 12 (Golomb's sequence)

Comments

Question 16 (OEIS A003215). The sequence

$$1, 7, 19, 37, 61, 91, 127, 169, 217, 271, 331, \ldots$$

is given by

$$a_n = 3n(n+1) + 1.$$

These numbers count lattice points inside a regular hexagon and are sometimes called the crystal ball sequence for the hexagonal lattice.

Question 18. The Golomb sequence is an integer sequence that grows slowly. For more information, see:

"The golden ratio from a calculus point of view," Mathematics Magazine, 2023.

`https://mat.uab.cat/web/matmat/wp-content/uploads/sites/23/2023/11/v2023n04.pdf`